Ronald V. Clarke

Situational Crime Prevention

ABSTRACT

Situational prevention seeks to reduce opportunities for specific categories of crime by increasing the associated risks and difficulties and reducing the rewards. It is composed of three main elements: an articulated theoretical framework, a standard methodology for tackling specific crime problems, and a set of opportunity-reducing techniques. The theoretical framework is informed by a variety of "opportunity" theories, including the routine activity and rational choice perspectives. The standard methodology is a version of the action research paradigm in which researchers work with practitioners to analyze and define the problem, to identify and try out possible solutions, and to evaluate and disseminate the results. The opportunity-reducing techniques range from simple target hardening to more sophisticated methods of deflecting offenders and reducing inducements. Displacement of crime has not proved to be the serious problem once thought, and there is now increasing recognition that situational measures may result in some "diffusion of benefits" to crimes not directly targeted.

More than ten years ago, situational prevention was defined in a *Crime and Justice* essay as "comprising measures directed at highly specific forms of crime that involve the management, design, or manipulation of the immediate environment in as systematic and permanent a way as possible so as to reduce the opportunities for crime and increase its risks as perceived by a wide range of offenders" (Clarke 1983, p. 225). Such measures include familiar forms of "target hardening" and property marking; more sophisticated technology including intruder alarms

Ronald Clarke is dean of the School of Criminal Justice at Rutgers—The State University of New Jersey. Thanks are due to several commentators on the draft of this essay, especially to Lawrence Sherman, University of Maryland.

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and electronic merchandise tags; the surveillance of specific locations provided by employees such as shop assistants and custodians and other attempts to capitalize on natural surveillance; exact fare systems, personal identification numbers for car radios, and a variety of other measures to reduce crime inducements; and some less easily categorized measures such as the use of public ordinances to control troublesome late-night entertainment spots and the separation of rival soccer fans into different enclosures at the stadium.

While the original definition requires little amendment (explicit mention would now also be made of reducing the rewards of crime), much else has changed. In particular, the theoretical basis of situational prevention has been strengthened with the development of routine activity and rational choice approaches, experience has accumulated of the successful application of the concept in a wide variety of crime contexts, a more detailed classification of situational measures has been developed, and the threat presented by displacement has receded as its limits have been clarified by research (Clarke 1992).

These achievements have helped to establish situational prevention in a number of European countries as one component in a broader array of crime control policies. Both in Holland and Great Britain, situational prevention is promoted by crime prevention units within central government. In Sweden, a semiautonomous governmental agency serves the same purpose. Given that the limitations are being increasingly recognized of alternative approaches to crime control, such as rehabilitation, deterrent sentencing, intensive policing, and incapacitation, it may be surprising that situational prevention has not been even more widely embraced by academics and policy makers. The lack of interest in the United States may be due partly to the disappointing results attending some early attempts to implement crime prevention through environmental design (CPTED), though Bright (1992) has noted the absence of any crime prevention policy in America, which he attributes to the dislike of government intervention, as well as to the strong ethos of individual responsibility that results in punishment being seen as the most appropriate response to law breaking. More generally, situational prevention conflicts with the mind-sets of criminologists and the vested interests of criminal justice practitioners. Many of the former seem unable or unwilling to make the paradigm shift involved in focusing on criminogenic situations rather than so exclusively on criminal actors, while for most of the latter there is scant appeal in a crime control strategy that depends so little on the

criminal justice system. The main reason for the relative lack of interest in situational prevention, however, may be unresolved difficulties about the implementation and delivery of situational measures and deeper concerns it provokes about the directions in which society may be moving with the approach of the twenty-first century (Bottoms 1990). Addressing these social concerns and the practical problems of implementation are among the challenges facing situational prevention in the forthcoming decade.

These implementation issues are discussed in Section V, at the end of this essay. In Section I, the theoretical background to situational prevention is described, and in Section II, a model of the opportunity structure is presented. In Section III, twelve opportunity-reducing techniques are described, together with some examples of their successful application. In Section IV, issues relating to displacement are discussed.

I. Theoretical Background

While the measures listed in the opening paragraph of this essay share the purpose of reducing opportunities for highly specific forms of crime and in this sense satisfy the definition of situational prevention, few were explicitly developed within a situational prevention framework. Rather, they were developed in isolation by people hard-pressed to solve crime problems encountered in their particular areas of responsibility. One of situational prevention's most important goals, therefore, second only to the objective of encouraging applications of the concept in fresh contexts, has been to provide a more scientific framework for some practical and commonsense thinking about how to deal with crime. This framework has three components, the first of which is a standard action research methodology (Lewin 1947) consisting of five sequential stages: (1) collection of data about the nature and dimensions of a specific crime problem, (2) analysis of the situational conditions that permit or facilitate the commission of the crimes in question, (3) systematic study of possible means of blocking opportunities for these particular crimes, including analysis of costs, (4) implementation of the most promising, feasible, and economical measures, and (5) monitoring of results and dissemination of experience (Gladstone 1980).

Possibly because of its extensive pedigree, the methodological component of situational prevention has been accepted with little change, but the two other components consisting of its theoretical underpinnings and the taxonomy of preventive measures are still undergoing

considerable development. The taxonomy is discussed in Section III, while this section describes the evolution of the theory supporting situational prevention. This description includes the origins of the theory in some Home Office research on institutional treatments, its relationship with work on CPTED and "defensible space" that had been developed contemporaneously in the United States, and the more recent infusion of concepts from rational choice, routine activity, and other "opportunity" theories.

A. Early Work by the Home Office Research Unit

The development of situational prevention was stimulated by the results of work on correctional treatments undertaken in the 1960s and 1970s by the Home Office Research Unit, the British government's criminological research department (Clarke and Cornish 1983). This work contributed to the demise of the rehabilitative ideal (Brody 1976) and forced researchers in the unit, charged with making a practical contribution to criminal policy, to review the scope and effectiveness of other forms of crime control. The review concluded that there was little prospect of reducing crime through the essentially marginal adjustments that were practically and ethically feasible in relation to policies of incapacitation, deterrent sentencing, preventive policing, or "social" prevention (Tilley 1993*a*). But it did identify reducing opportunities for crime as a worthwhile topic for further research, largely on the basis of some findings about misbehavior in institutions. It had been discovered in the course of work on rehabilitation that the probability of a youth's absconding or reoffending while resident in a probation hostel or training school seemed to depend much more on the nature of the institutional regime to which he was exposed than on his personality or background (Tizard, Sinclair, and Clarke 1975). Particularly important appeared to be the opportunities for misbehavior provided by the institutional regime—opportunities that could be "designed out."

If institutional misconduct could in theory be controlled by manipulating situational factors, it was reasoned that the same might be true of other, everyday forms of crime. Though this was not consistent with most current theory (which treated crime principally as the expression of biological inheritance, personality, or social factors), support for the Home Office position was found in criminological studies that had found immediate situational influences to be playing an important role in crime. These included Burt's (1925) studies of delinquency

in London, showing that higher rates of property offending in the winter were promoted by longer hours of darkness; Hartshorne and May's (1928) experimental studies of deceit, showing that the likelihood of dishonest behavior by children was dependent on the level of supervision afforded; geographical studies showing that the distribution of particular crimes is related to the presence of particular targets and locations such as business premises, drinking clubs, and parking lots (Engstad 1975); and demonstrations that fluctuations in auto theft reflect the number of opportunities as measured by the numbers of registered vehicles (e.g., Wilkins 1964).

The Home Office position was also consistent with some psychological research on personality traits and behavior that was finding a greater-than-expected role for situational influences (Mischel 1968) and with an emerging body of work on the sociology of deviance, including studies by Matza (1964), who argued against deep motivational commitment to deviance in favor of a "drift" into misconduct; by Briar and Piliavin (1965), who stressed situational inducements and lack of commitment to conformity; and by Yablonsky (1962) and Short and Strodtbeck (1965), who demonstrated the pressures to deviance conferred by working-class gang membership.

Taken together, this body of work suggested that criminal conduct was much more susceptible to variations in opportunity and to transitory pressures and inducements than conventional "dispositional" theories allowed. It was also becoming clear from interviews with residential burglars (Scarr 1973; Reppetto 1974; Brantingham and Brantingham 1975; and Waller and Okihiro 1978) that the avoidance of risk and effort plays a large part in target selection decisions. This dynamic view of crime provided a more satisfactory basis for situational prevention and led to the formulation of a simple "choice" model (Clarke 1977, 1980). This required information not only about the offender's background and current circumstances but also about the offender's immediate motives and intentions, moods and feelings, moral judgments regarding the act in question, perception of criminal opportunities and ability to take advantage of them or create them, and assessment of the risks of being caught as well as of the likely consequences.

This model, dubbed "situational control theory" by Downes and Rock (1982), was subsequently developed into the rational choice perspective on crime (see below), but it served initially to deflect criminological criticism of the atheoretical nature of situational prevention and,

more important, to guide thinking about practical ways of reducing opportunities for crime.

B. Defensible Space, CPTED, and Problem-oriented Policing

Preceding situational prevention by a few years were two related strands of policy research in the United States, "defensible space" (Newman 1972) and "crime prevention through environmental design" or CPTED (Jeffery 1971). Neither had provided the stimulus for its development, though arguments and examples that both provided were useful in its promotion. Nor did CPTED or defensible space contribute materially to the theoretical basis of situational prevention. Both were more narrowly focused on the design of buildings and places (whereas situational prevention seeks to reduce opportunities for crime in all behavioral contexts), and both embraced somewhat questionable theories of human behavior. In Jeffery's (1971) case, his "biosocial" approach included not merely environmental change but also the more controversial ingredient of treating the genetic basis of crime. In the case of Newman (1972), his "defensible space" concept was influenced not only by Jacobs's (1961) ideas about the relationship between crime and the layout of American cities, but also by some debatable ethological ideas (Mayhew 1979) about human "territoriality" developed by authors such as Ardrey (1966).

The work on "defensible space" was invaluable, however, in providing a powerful tool for analyzing problems of crime and vandalism in public housing in Britain. Newman was especially critical of huge buildings which made it impossible for residents to recognize strangers, numerous unsupervised access points that made it easy for offenders to enter projects and escape, the location of projects in high-crime areas, and the projects' stigmatizing appearance. His work included a wealth of detailed design suggestions for creating "defensible space" through reducing anonymity, increasing surveillance, and reducing escape routes for offenders. One of the first situational prevention projects undertaken by the Home Office involved testing Newman's ideas in a British context (Wilson 1978). While its results were not wholly supportive of "defensible space" concepts, it was succeeded by more comprehensive studies (Coleman 1985) and undoubtedly contributed to British government interest in designing more secure and livable public housing environments (Burbidge 1984).

If situational prevention rode on the coattails of defensible space, it has more recently been hobbled by its association in the United States

with CPTED, partly because of the latter's biological components, but more importantly because of the failure of some federally sponsored CPTED projects undertaken by the Westinghouse Corporation. These were massively funded, but misconceived because they attempted to extend the defensible space concept to school and commercial sites where "territorial" behavior is much less natural than in the residential context (see Jeffery 1977). Partly as a result of these failures, both government funding agencies and criminologists have shown little recent interest in CPTED, and, with the exception of some studies undertaken by Jeffery and his associates of convenience store robbery (Hunter and Jeffery 1992) and some work in Canada by Brantingham and Brantingham (1988), little CPTED research has been published in North America in the last decade.

More recent interest has greeted conceptually related (Hope 1994) though independent work on "problem-oriented policing" by Goldstein (1979, 1990), who has argued that the route to greater operational effectiveness for the police is not through improvements in organization and management but through the devising of tailor-made solutions for the range of problems that they are called on to handle. Goldstein recognized the need for evaluation, and his formulation of problem-oriented policing appears to reflect the same action research paradigm underpinning situational prevention (see Goldstein 1990, p. 103). Nevertheless, some important differences exist between the concepts. In particular, problem-oriented policing is a management approach designed to make most efficient use of police resources (Moore 1992), while situational prevention is a crime control approach open not just to the police but to any organizational or management structure.

C. The Rational Choice Perspective

The earlier "choice" model formulated to guide situational prevention efforts has more recently been developed into a "rational choice" perspective on crime (Clarke and Cornish 1985; Cornish and Clarke 1986). This borrows concepts from economic theories of crime (e.g., Becker 1968) but seeks to avoid some of the criticisms made of these theories, in particular the following: (i) that under economic models the rewards of crime are treated mainly in material terms (how much money an offender can make), while mostly ignoring rewards that cannot easily be translated into cash equivalents; (ii) that economists have not been sensitive to the great variety of behaviors falling under the general label of crime, with their variety of costs and benefits, and

instead have tended to lump them together as a single variable in their equations; (iii) that the formal mathematical modeling of criminal choices by economists often demands data that are unavailable or can only be pressed into service by making unrealistic assumptions about what they represent; and, finally, (iv) that the economist's image of the self-maximizing decision maker, carefully calculating his or her advantage, does not fit the opportunistic and reckless nature of much crime (Clarke and Felson 1993).

Under the new formulation of the rational choice perspective, relationships between concepts were expressed, not in mathematical terms as was the case in Becker's normative model, but in the form of "decision" diagrams (Clarke and Cornish 1985; Cornish and Clarke 1986). Concepts were adapted from the other disciplines involved in the analysis of criminal decision making, as well as economics, to give greater weight to noninstrumental motives for crime and the "limited" nature of the rational processes involved. It was assumed, in other words, that crime is purposive behavior designed to meet the offender's commonplace needs for such things as money, status, sex, and excitement, and that meeting these needs involves the making of (sometimes quite rudimentary) decisions and choices, constrained as these are by limits of time and ability and the availability of relevant information.

A second important new premise was that a decision-making approach to crime requires that a fundamental distinction be made between criminal involvement and criminal events. (This parallels the distinction made by Gottfredson and Hirschi [1990] between criminality and crime.) Criminal involvement refers to the processes through which individuals choose to become initially involved in particular forms of crime, to continue, and to desist. The decision processes at each of these stages are influenced by a different set of factors and need to be separately modeled. In the same way, the decision processes involved in the commission of a particular crime (i.e., the criminal event) are dependent on their own special categories of information. Involvement decisions are characteristically multistage and extend over substantial periods of time. Event decisions, by contrast, are frequently shorter processes, using more circumscribed information largely relating to immediate circumstances and situations.

Finally, and this is of special importance for situational prevention, it was assumed that the decision processes and information used will vary greatly between different offenses. To ignore these differences, and the situational contingencies associated with them, may be to re-

duce significantly the scope for intervention. Nor will it be enough to make the usual crude distinctions between crime categories, such as that between commercial and residential burglary. Thus Poyner and Webb (1991) have shown that residential burglaries committed for cash and jewelry show a quite different pattern in British communities than do those committed for electrical goods. The former seemed to be committed primarily by opportunistic offenders working on foot, while the latter seemed to rely on the use of a car. These differences in modus operandi have implications for prevention; Poyner and Webb suggested, in particular, that to prevent burglaries of electrical goods houses should face each other across the street and overlook access roads, while to prevent burglaries of cash and jewelry cover at the front of houses in the form of shrubs or fences should be removed.

While the new formulation of the rational choice perspective was primarily developed to assist thinking about situational prevention, it was not intended to be limited to this role. Indeed, Cornish (1993) has argued that many features of the rational choice perspective make it particularly suitable to serve as a criminological "metatheory" with a broad role in the explanation of a variety of criminological phenomena.

D. Environmental Criminology, Routine Activities, and Lifestyles

Rational choice premises have generally been supported by recent studies in which offenders have been interviewed about motives, methods, and target choices. The offenders concerned have included burglars (e.g., Walsh 1980; Maguire 1982; Bennett and Wright 1984; Nee and Taylor 1988; Biron and Ladouceur 1991; Cromwell, Olson, and Wester Avary 1991), shoplifters (Walsh 1978; Carroll and Weaver 1986), muggers (Lejeune 1977; Feeney 1986), and bank and commercial robbers (New South Wales Bureau of Crime Statistics and Research 1987; Kube 1988; Nugent et al. 1989). These studies of offender decision making constitute one of two major analytic paths followed in the past decade by "environmental criminology" (Brantingham and Brantingham 1991). The other path has involved "objective analysis of the spatial and temporal variation in crime patterns in order to discover aggregate factors influencing the patterns" (Brantingham and Brantingham 1991, p. 239). Much of this work, which has greatly expanded evidence about situational factors in crime, consists of macro analyses of climatic variations or city structure. One important component has been research on the criminal's "journey to work," an example of which is Poyner and Webb's (1991) study, mentioned above, with its implica-

tions for preventing different forms of residential burglary. Other similar studies by Brantingham and Brantingham (1975), Maguire (1982), and Rengert and Wasilchick (1985) have shown, among other things, that the risks of commercial robbery may be increased by being located close to a main road, and those of residential burglary by being located on the outskirts of an affluent area, in both cases because the offender's target search time is thereby reduced.

Research on the criminal's journey to work is conceptually related to another body of criminological work—routine activity theory—which has also contributed to the theoretical base of situational prevention. The routine activity approach stated three minimal elements for direct-contact predatory crime: a likely offender, a suitable target, and the absence of a capable guardian against crime (Cohen and Felson 1979). It avoids speculation about the source of the offender's motivation, which distinguishes it immediately from most other criminological theories. Instead, it focuses on the convergence in space and time of the three elements of crime, that is to say, on the conditions favoring the occurrence of a criminal event, rather than the development of a criminal disposition. This reflects its intellectual roots in the human ecology of Amos Hawley (1950), who recognized that the timing of different activities by hour of day and day of week was important for the understanding of human society. These points were central also to the routine activity approach. No matter at what level data were measured or analyzed, that approach kept returning to specific points in time and space and to changes from moment to moment and hour to hour in where people are, what they are doing, and what happens to them as a result (Clarke and Felson 1993). In support of their approach, Cohen and Felson (1979) sought to demonstrate that increases in residential burglary in the United States between 1960 and 1970 could largely be explained by changes in "routine activities" such as the increasing proportion of empty homes in the day (due to more single-person households and greater female participation in the labor force) and the increased portability of televisions and other electrical goods.

Cohen and Felson's analysis also illustrates the relationship between routine activity theory and some victimological work on "lifestyles," stimulated by the flood of National Crime Survey data first released in the 1970s (Hindelang, Gottfredson, and Garofalo 1978). One of the tenets of "lifestyle" theory is that the differential risks of victimization are partly a function of differential exposure to motivated offenders (Fattah 1993). This exposure varies not only with the sociodemo-

graphic characteristics of the victim (age, race, place of residence, etc.) but also with the victim's lifestyle. A person's work and leisure activities that increase exposure to potential offenders (such as alcohol consumption in public places or late-night use of public transport) increase the risks of victimization. The implication of this is that risks might be reduced by modifying patterns of activity. A further important finding of victimological research, the implications of which are being explored in a series of recent studies by Ken Pease and colleagues (see Farrell and Pease [1993] for a review), is that some people and targets are repeatedly subject to victimization and might therefore be prime candidates for preventive attention (see Farrell, in this volume). A similar point has been made by Sherman, Gartin, and Buerger (1989) in relation to the "hot spots" of crime, places that are the source of repeated calls for assistance to the police.

Lifestyle and routine activity theories have both made opportunity a respectable topic of research in criminology and helped situational prevention attract serious scholarly interest. Both approaches are still evolving, and Felson himself has made some attempts to expand the scope of routine activity theory. He has defined minimal elements for some categories of crime other than direct-contact predatory offenses (Felson 1992) and, in order to accommodate social control theory (Hirschi 1969), has proposed a fourth minimal element for predatory crimes, "the intimate handler," or someone who knows the likely offender well enough to afford a substantial brake on the latter's activities (Felson 1986). Clarke (1992) has argued that the contribution of routine activity theory to crime prevention could be enhanced by adding a fifth element which he refers to as "crime facilitators." These are such things as automobiles, credit cards, and weapons that constitute the essential tools of particular forms of crime.

II. The Opportunity Structure for Crime

Environmental criminology, the rational choice perspective, and routine activity and lifestyle theories have all helped to strengthen situational prevention in different ways, reflecting their different origins and the purposes for which they were developed. By interviews with offenders and analyses of crime patterns, environmental criminology has provided rich information about the motives and methods of offenders, which has been valuable in thinking about countermeasures. The rational choice perspective has provided a framework under which to organize such information so that individual studies produce more

general benefits and has also assisted analysis of displacement. Lifestyle theory has focused attention on what victims might do to reduce their risks of crime. And routine activity theory has served to extend preventive options by directing attention to features of the three essential elements of crime and their convergence. For example, the idea of convergence has led to the suggestion that "deflecting offenders" be recognized as a distinct technique of situational prevention (Barr and Pease 1990; Clarke 1992).

Cusson (1986) and Felson (forthcoming) have argued that the differences among the various theoretical approaches may turn out to be mainly of historical interest and that a synthesis is inevitable and desirable. The model of the opportunity structure for crime discussed in this section represents one such attempt at integration.

A. The Concept in Outline

Under this model (see fig. 1), which includes the dispositional variables of traditional criminology as well as the situational ones of the newer theories, there are three components of the criminal opportunity structure. These are targets (cars, convenience stores, automated teller machines, etc.), victims (e.g., women alone, drunks, strangers), and crime facilitators. These latter include tools, such as guns and cars, as well as disinhibitors, such as alcohol or other drugs.

The supply of targets and their nature is a function of, first, the physical environment, including the layout of cities, the kinds of housing, technology and communications, transportation and retailing systems, the numbers of vehicles and the supply of drugs and alcohol, and, second, the lifestyles and routine activities of the population, including patterns of leisure, work, residence, and shopping. These patterns either hinder or facilitate guardianship. The physical environment also determines the supply of facilitators, while lifestyles and routine activities play a large part in supplying the victims of personal and sexual attacks. Physical environment and lifestyles and routine activities are themselves determined by the broader socioeconomic structure of society, including demography, geography, urbanization and industrialization, health and educational policy, and legal and political institutions. The numbers of potential offenders and their motives are also determined partly by the socioeconomic structure of society through many of the mechanisms (alienation, subcultural influence,

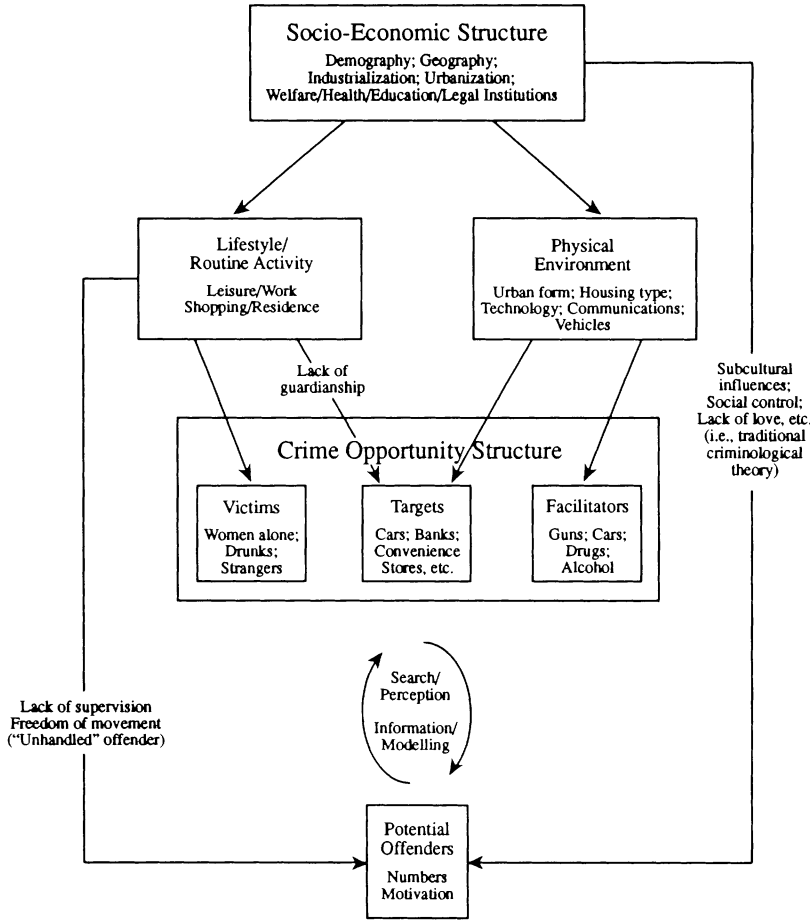

FIG. 1.—The opportunity structure for crime

neglect, and lack of love, etc.) identified by traditional criminology and partly by lifestyle and routine activities which affect the nature of social control afforded by "intimate handlers," and in other ways.

The opportunity structure is not simply a physical entity, defined at any one time by the nature of the physical environment and the routine activities of the population. Rather, a complex interplay between potential offenders and the supply of victims, targets, and facilitators determines the scale and nature of opportunities for crime. Potential offenders learn about criminal opportunities from their peers, the media, and their own observation, but they are differentially sensi-

tized to this information, as well as being differentially motivated to seek out and create opportunities (Maguire 1980; Bennett and Wright 1984). Thus offender perceptions and judgments about risks, effort, and rewards play an important part in defining the opportunity structure.[1] These judgments also play a determining role at the subsequent stage of crime commission, where figure 1 stops short.

B. Policy Implications

Little account is taken in figure 1 of society's efforts to control and prevent crime because this might have diverted attention from the purpose of integrating the various opportunity perspectives. Some brief discussion is needed, however, of questions about the scope and reach of situational prevention which arise from consideration of the model of the opportunity structure. The first of these can be phrased as follows: if everything seems to flow from the socioeconomic structure, why not focus preventive effort at that level? Large-scale reductions in a wide range of crimes might be achieved by tackling the disposition to offend at its roots, through improved welfare and educational programs. This would appear to be more efficient than the undertaking of a vast number of small-scale efforts to address highly specific crime problems that is implied by the situational focus.

One answer to this is that "social" forms of crime prevention are already focused at the socioeconomic level and that the opportunity structure requires attention in its own right. A second, more radical reply has been provided by Morris and Hawkins (1970) and Wilson (1975), who have noted the great difficulty of bringing about some social changes (e.g., how parents can be made to love their children more), as well as that decisions about other change will only be marginally influenced by arguments about crime: most of the changes that might bring about reductions in crime, such as better welfare and education, are usually seen as desirable but as demanding resources that society cannot afford.

A second question that concerns the differential balance between dispositional and situational factors for various crimes has been raised by Trasler (1986, 1993), among others, and is as follows: even if situational measures may be appropriate in dealing with certain categories of crime (especially opportunistic property offenses), is it not the case

[1] Although not made explicit in fig. 1, it is recognized that these perceptions and judgments are not always individual activities, but sometimes will be collaborative with peers and co-offenders.

that more deeply motivated crimes committed by desperate or hardened offenders can only be prevented by addressing fundamental dispositional causes? This can be answered by the argument that all classes of crime, even those most deeply motivated by anger or despair, are greatly affected by situational contingencies. Whether an assault becomes a homicide may depend on the presence at the scene of a lethal weapon (Cook 1991); similarly, whether someone commits suicide is importantly determined by the availability of an acceptable method (Clarke and Lester 1989). With respect to crimes thought to be the province of "hardened" offenders, evidence is now accumulating of successes achieved by situational prevention, including the virtual elimination of aircraft hijackings by baggage screening (Wilkinson 1986) and substantial reductions in robbery achieved by target-hardening measures in post offices (Ekblom 1988), convenience stores (Hunter and Jeffery 1992), and banks (Gabor 1990; Grandjean 1990; Clarke, Field, and McGrath 1991).

Heal and Laycock (1986) and Gabor (1990) have nevertheless maintained that, because many violent crimes are less common and less likely to cluster in time or space, they are also less amenable to situational controls. However, this may be true only to the extent that the measures considered are traditional ones of target hardening and surveillance. In Section III some examples are provided of the successful control of violence through deflecting offenders (e.g., by preventing the congregation of large groups of drunken youths at pub closing time) or by controlling crime facilitators such as alcohol and guns. One instructive example is provided by the introduction of Caller-ID in New Jersey, which, by threatening the anonymity of callers, seems to have produced a substantial reduction in obscene phone calling (Clarke 1990). Without this evidence, many people might have argued that obscene phone calling, a sexual crime that seems to strike at random, is precisely the kind of offense that would be unamenable to situational controls. A similar argument might have been made about domestic violence, but encouraging evidence is beginning to emerge from an experimental program in England that providing personal alarms to repeat victims may inhibit the aggressor (Farrell and Pease 1993). The lesson is that the limits of situational prevention should be established by closely analyzing the circumstances of highly specific kinds of offense, rather than by theoretical arguments about the presumed nature of motives for broad categories of crime such as sexual or violent offenses.

A third question concerns deterrence. It relates specifically to the issue of offender choice and is as follows: rather than attempting to manipulate the opportunity structure (with the attendant costs and inconvenience of this strategy), might it not be more efficient simply to raise the stakes of offending by heavier punishments? The answer to this is that most interviews with offenders have shown that they pay much closer attention to the immediate chances of getting caught than to the nature of the punishment they might receive later. It is more efficient, therefore, to make the offender more fearful of being caught than of being subsequently punished, and one component of situational prevention does indeed consist of increasing the risks of being caught (a process referred to as "situational deterrence" by Cusson [1993]).

A final set of questions concerns the interplay between the objective reality of the opportunity structure and the way this is perceived by potential offenders. How do offenders learn about criminal opportunities, and what factors come into play when they make decisions about which ones to pursue? What proportion of crimes are the result of opportunities seized, and what proportion are the result of opportunities that are sought or created (Maguire 1980; Bennett and Wright 1984)? At issue here is the question of whether opportunities for crime really are in infinite supply, as some have argued. If so, this has serious implications for a strategy advocating their reduction. What matter the few reductions that can be achieved in criminal opportunities if these are infinite?

Consideration of the realities of crime may supply an answer to this question. While it may be true that every dwelling and automobile provides in theory not just one opportunity for crime, but if considered over time, a set of almost endless opportunities, this overlooks the substantial guardianship afforded dwellings and automobiles for much of the time (Clarke 1984). Even when unguarded, they may in actuality provide few rewards for crime. The average dwelling contains only a few portable goods that can be converted into cash, and there are limits to the number of stolen VCRs and television sets that the offender can store. It is also unclear how many such "hot" items can be unloaded onto the market without provoking a determined response from law enforcement. Clarification of these issues needs to be sought in more research of the kind recently published by Cromwell, Olson, and Wester Avary (1991) in which they undertook detailed interviews with residential burglars about their working methods.

A more basic program of research into the ways that behavior is routinely channeled by society in order to protect the privacy and security of individuals and institutions would also be of value. This could explore a variety of questions such as the following.

1. How much geographical space in a town or city has not been privatized and is readily accessible to strangers?
2. What proportion of goods for sale in retail establishments could or could not easily be stolen by shoppers?
3. What are the subtle cues in private places that progressively reduce access with the degree of intimacy between "owners" and "visitors"? (You might invite me into your living room, but not into your bedroom.)
4. What limits an offender's willingness to travel?

Answers to these questions would assist discussion about the supply of crime targets. A similar, but a possibly more restricted set of questions needs to be asked about facilitators. For example, much more needs to be learned about the reasons guns are used in committing crimes in some cultures but not in others with similar rates of crime, even when guns are equally available (Tonso 1982; Harding 1993). Some clues are provided by Harding (1993) who has found that robbers in Western Australia who had been introduced to guns by their fathers or other authority figures were much less likely to use guns in their robberies than those introduced to guns by peers (see also Wright and Rossi [1982] for similar findings in the United States). Harding also reports that, even though murder rates for Australian Aboriginals are very high, the rates of gun murder are relatively low. As one offender interviewed by Harding (1993, p. 94) explained the reason, "Guns are for shooting tucker (i.e., food) not people."

III. Opportunity-reducing Techniques

Putting situational prevention on a more rigorous footing has involved not only the development of a theoretical framework and the formulation of a standard methodology but also various efforts to classify the opportunity-reducing techniques that have been identified. None of these classifications has been widely used, however, which seems to be due to a continually expanding repertoire of measures, resulting from fresh applications of situational prevention, and improved theoretical understanding of the relationship between situational vari-

ables and crime. For example, Clarke's (1992) suggestion mentioned above that "deflecting offenders" should be a discrete category of opportunity-reducing measures arose from a consideration of routine activity theory.

This suggestion was made in the context of developing a twelve-category classification of techniques that builds on a more limited classification proposed earlier by Clarke and Mayhew (1980). The twelve categories of the new classification are described below (and presented in summary form in table 1) with some successful examples of the application of each technique.

Before launching into this description, however, some brief points should be made about the examples of "success." There are a large number of these (and more might have been cited [see Poyner 1993]) relating to a wide variety of settings and crime types. Taken as a whole, there can be little question that they provide credible evidence of significant crime reduction effects. Nevertheless, it has to be recognized that in most cases the individual evaluations were comparatively rudimentary. Follow-ups were often short, so little is known about the durability of success. True experimental designs were almost completely absent from the studies (most of which consisted of simple time-series or quasi-experimental designs), with the result that in most cases it was impossible to be sure that the identified situational measure had produced the observed reduction in crime.[2] These reductions might have been due instead to some other contemporaneous, unmeasured change or even from regression to the mean.[3] In many of the studies, several preventive measures had been deployed at the same time, and it was impossible to be sure which combination, if any, was responsible for the success. Moreover, for reasons discussed at greater

[2] Experimental designs are difficult to undertake in field settings and, even when successfully completed, can be hard to interpret. For example, Farrington et al. (1993) have recently reported an experimental study of shoplifting prevention, in which three measures (electronic tagging, store redesign, and security guards) were systematically compared for their effectiveness. They concluded that the security guards were ineffective, but with the caveat that this may have been due to the layout of the store, which prevented adequate oversight of customers, or to the inexperience, unimpressive physiques, and lack of training of the particular guards concerned. It is doubtful that the testing of rival hypotheses such as these within the confines of a rigorous experimental design would often be a practicable proposition, and, indeed, it seems preferable in the present state of knowledge to undertake as many evaluations as possible while compensating for weak designs with detailed observation of the process of implementation. (The value of this is illustrated by Farrington et al.'s observations about the caliber of the security guards in their shoplifting research.)

[3] For an illustration of this problem, see Sherman's (1992) discussion of the successes claimed in preventing robbery in convenience stores through employing two clerks.

TABLE 1

The Twelve Techniques of Situational Prevention

Increasing the Effort	Increasing the Risks	Reducing the Reward
1. Target hardening:	5. Entry/exit screening:	9. Target removal:
Steering locks	Baggage screening	Removable car radio
Bandit screens	Automatic ticket gates	Exact change fares
Slug rejector device	Merchandise tags	Phonecard
2. Access control:	6. Formal surveillance:	10. Identifying property:
Fenced yards	Security guards	Property marking
Entry phones	Burglar alarms	Vehicle licensing
ID badges	Speed cameras	Personal identification numbers for car radios
3. Deflecting offenders:	7. Surveillance by employees:	11. Removing inducements:
Tavern location	Park attendants	Graffiti cleaning
Street closures	Pay phone location	Rapid repair
Graffiti board	Closed-circuit television systems	"Bum-proof" bench
4. Controlling facilitators:	8. Natural surveillance:	12. Rule setting:
Gun controls	Street lighting	Customs declaration
Credit card photo	Defensible space	Income tax returns
Caller-ID	Neighborhood watch	Hotel registration

SOURCE.—Adapted from Clarke (1992).

length below, it was rarely possible to be sure that displacement had not negated the benefits of the observed reductions in crime, although in many cases there was little prima facie evidence of this. All of this means that, while the weight of the positive evidence cannot convincingly be denied, the successes claimed in any individual study may be open to question.

A. Target Hardening

The most obvious way of reducing criminal opportunities is to obstruct the vandal or the thief by physical barriers through the use of locks, safes, screens, or reinforced materials.

The introduction of steering locks on both new and old cars in West Germany in 1963 produced a substantial decline in the rate of car theft for the country that has persisted to this day (Webb and Laycock 1992; Webb 1994).

Decker's (1972) study showed that the use of slugs in New York parking meters was substantially reduced by changes in design, including a slug-rejector device.

The fitting of transparent screens to shield the bus driver significantly reduced assaults on one transit system (Poyner et al. 1988), and the installation of antibandit screens on post office counters in London in the early 1980s was conservatively estimated by Ekblom (1988) to have cut robberies by 40 percent.

Challinger (1991) identifies strengthened coin boxes as a significant factor in reducing incidents of deliberate damage to public telephones in two Australian states—South Australia and the Northern Territory—from a peak of nearly 6,000 in 1988 to just over 1,100 one year later.

Clarke, Field, and McGrath (1991) attributed marked reductions in armed robbery of Australian banks during the late 1980s to a program of target hardening, including the installation of fixed and "pop-up" bulletproof screens.

B. Access Control

Access control refers to measures intended to exclude potential offenders from places such as offices, factories, and apartment buildings. The portcullises, moats, and drawbridges of medieval castles suggest its preventive pedigree may be as lengthy as that of target hardening. It is also a central component of defensible space, arguably the start of scientific interest in situational prevention. A sophisticated form of

access control lies in the use of electronic personal identification numbers (PINs) that are needed to gain access to computer systems and bank accounts.

Matthews (1990) has shown that a road closure scheme to reduce access by cruising "johns" was one of the measures contributing to the rehabilitation in 1984 of a red-light district in Finsbury Park, an area of North London. Not only was prostitution very significantly reduced but also a wide range of crimes declined from a total of 475 in the twelve months prior to implementation of the scheme to 275 in the twelve months after.

Poyner and Webb (1987) found that a combination of access controls introduced on a South London public housing estate, including entry phones, fencing around apartment blocks, and electronic access to the parking garage, achieved a significant reduction in vandalism and theft. They also found that the introduction of a reception desk on the ground floor of a high-rise building achieved a marked reduction in vandalism, graffiti, and other incivilities.

C. Deflecting Offenders

The provision of public urinals, litter bins, and "graffiti boards" (the latter being supplied for people's public messages) channels people's behavior in more acceptable directions and constitutes an example of deflecting offenders. In Britain, arrival and departure of soccer fans has been scheduled to avoid long periods of waiting around that promote trouble, and rival fans have been physically separated within the stadium to reduce the probability of fights (Clarke 1983). Avoiding the concentration of licensed premises in particular parts of the city has been proposed by Hope (1985) as a way of avoiding the closing-time brawls that result when large crowds of drunken people spill onto the streets late at night. Finally, Shearing and Stenning (1984) provide a fascinating glimpse into the ways in which sophisticated crowd control and management—involving the use of pavement markings, signs, physical barriers (that make it difficult to take a wrong turn), and instructions from cheerful Disney employees—greatly reduce the potential for crime and incivility in Disneyland.

Bell and Burke (1989) show that the leasing of a downtown parking lot in Arlington, Texas, relieved severe congestion on weekend nights in nearby streets, and associated crime problems, by providing a venue for teenage cruising.

Poyner and Webb (1987) show that thefts from shopping bags at

market stalls in Birmingham, England, were reduced by over 70 percent within two years by reducing congestion, which increased the difficulty of pickpocketing and other "stealth" thefts.

D. Controlling Facilitators

Saloons in the Wild West routinely required their clienteles to surrender their weapons on entry because of the risk of drunken gun fights. In more recent times the manufacture of "less lethal weapons" in the form of guns that shoot wax bullets, electricity, or tranquilizers has been advocated (Hemenway and Weil 1990). The Scottish Council on Crime (1975) suggested that in some pubs beer should be served in plastic mugs to prevent their use as weapons, and recent studies in Britain of the injury potential of different kinds of broken glass have led to the recommendation that toughened glass be used for beer glasses (Shepherd and Brickley 1992). Controls on a range of other crime facilitators have been proposed including alcohol (which facilitates almost all crime), automobiles (which facilitate crimes such as burglary), checks and credit cards (which facilitate fraud), and telephones (which may facilitate drug dealing, frauds, and sexual harassment). Licenses and IDs restrict access to cars and alcohol for juveniles, the most crime-prone group. Access to automobiles by intoxicated drivers can be restricted by the use of Breathalyzers built into the ignition, a measure that is now being mandated for some recidivist drunk drivers (Jones and Wood 1989; Morse and Elliott 1990). To reduce drug dealing, some telephone companies have removed public telephones from places where drug dealers congregate or have altered the phones so they cannot receive incoming calls (*Law Enforcement News* 1990).

Bjor, Knutsson, and Kuhlhorn (1992) argue that "rationing" the amount of alcohol that revelers were allowed to bring into a Swedish resort town on Midsummer Eve made an important contribution to reduced levels of drunkenness and disorderly conduct.

Knutsson and Kuhlhorn (1981) have shown that the introduction of identification procedures in Sweden produced a dramatic decline in the number of reported check frauds, from a peak of 15,817 in 1970 to 10–20 percent of these figures in later years.

Clarke (1990) has provided evidence that the introduction in New Jersey of Caller-ID, a service which allows the person answering to read the number of whoever is making the call, appears to have resulted in a substantial reduction, by about 25 percent, of obscene and annoying telephone calls.

E. Entry/Exit Screening

Entry screening differs from access control in that the purpose is less to exclude potential offenders than to increase the risk of detecting those who are not in conformity with entry requirements. These requirements may relate to prohibited goods and objects or, alternatively, to possession of tickets and documents. Exit screens serve primarily to deter the illegal removal of objects. Developments in electronics have resulted in the increasing use of these situational techniques in retailing, as evidenced by the spread of merchandise tagging, bar coding, and "electronic point of sales" systems (Hope 1991).

The introduction of baggage and passenger screening at most major airports in the world during the early 1970s contributed to a precipitate reduction in the number of airline hijackings from about seventy per year (Wilkinson 1977, 1986; Landes 1978).

Scherdin (1986) reports that the installation of book detection screens, similar to those found in thousands of libraries, reduced thefts of both books and audiovisual materials at one University of Wisconsin library by more than 80 percent.

DesChamps, Brantingham, and Brantingham (1991) provide evidence that the redesign of tickets to facilitate their inspection on Vancouver ferries produced a two-thirds reduction in fare evasion.

According to Clarke (1993), the installation of automatic ticket gates on the sixty-three central zone stations of the London Underground resulted in a two-thirds reduction of fare evasion throughout the system. He also concluded that the increase in fares collected should pay for the gates within three years of installation.

F. Formal Surveillance

Formal surveillance is provided by police, security guards, and store detectives, whose main function is to furnish a deterrent threat to potential offenders. Their surveillance role may be enhanced by electronic hardware, for example, by burglar alarms and closed circuit television (CCTV). Ways of enhancing police surveillance by enlisting the help of the public are also continually being expanded. Informant hot lines and "crime stopper" programs are two recent examples. "Curfew decals" on automobiles (Clarke and Harris 1992*a*) that indicate to patrolling police that the vehicle is not normally in use late at night represent a program directed specifically at auto theft prevention.

Rates of vandalism, assault, and fare dodging on subways and trams in three Dutch cities were substantially reduced by hiring 1,200 unem-

ployed young people to serve as safety, information, and control inspectors ("VICs") (van Andel 1989).

A problem-oriented policing initiative directed against auto theft in a large shipyard parking lot, featuring intensive surveillance of the yard by police officers, reduced thefts by more than one-half, or by an estimated 450 incidents in a sixteen-month period (Eck and Spelman 1988).

When CCTV was installed for the use of security personnel at a university's parking lots, Poyner (1991) found a substantial reduction in thefts, from ninety-two in the year before intervention to thirty-one in the year after. Tilley (1993*b*) has reported similar results for a number of other car parks in Britain, though in some of the cases he cites the CCTV was not monitored by security personnel.

Systematic, daily counting by security personnel of items of high-risk merchandise, such as VCRs and camcorders, resulted in declines of between 80 percent and 100 percent in the numbers of units stolen by employees at four large electronics stores and the associated distribution center in New Jersey (Masuda 1992).

Bourne and Cooke (1993) show that the widespread deployment of photo radar in Victoria, Australia, was the major factor in substantially reduced levels of speeding in 1991–92, with a consequent reduction of 25 percent in collisions, 40 percent in injuries, and 45 percent in fatalities.

Also in Australia, Homel (1993) has reported that the introduction of random breath testing (RBT) in New South Wales in 1982 was followed by an instantaneous 22 percent decline in total fatal crashes and a drop of 36 percent in alcohol-related fatal crashes relative to the previous three years. These declines have persisted as a result of a continued high level of RBT enforcement.

G. *Surveillance by Employees*

In addition to their primary function, some employees, particularly those dealing with the public, also perform a surveillance role. They include shop assistants, hotel doormen, park keepers, parking lot attendants, and train conductors. Canadian research has shown that apartment blocks with doormen are less vulnerable to burglary (Waller and Okihiro 1978). In Britain, less vandalism has been found on buses with conductors (Mayhew et al. 1976) and on public housing estates with resident caretakers (Department of Environment 1977). Public telephones in Britain sited where they get some surveillance by employees,

such as in pubs or railway stations, also suffer fewer attacks (Markus 1984).

Hunter and Jeffery (1992) report that ten out of fourteen studies they reviewed found that having two clerks on duty, especially at night, was an effective robbery prevention measure.

Comparison of pre- and post-twelve-month figures showed that the employment of attendants to cover high-risk periods of the day was followed by a two-thirds reduction in vehicle-related offenses at a parking lot in Basingstoke, England (Laycock and Austin 1992).

The installation of CCTV for the use of station staff on four high-risk stations on the London Underground was found to have produced substantial reductions in muggings and thefts (Mayhew et al. 1979).

Vandalism of seats on a fleet of double-deck buses in the North of England was substantially reduced through the provision of CCTV for drivers. Only a few buses were fitted with CCTV, but the effect was general, so that seat repairs for the fleet declined from a peak of about eighty per month to a third of that number subsequently (Poyner 1988).

Rewarding cashiers for detection of forged or stolen credit cards helped to reduce losses resulting from fraudulent use of cards at the four large stores of an electronics retailer in New Jersey from $1.1 million in 1991 to about $0.2 million in the first eleven months of 1992 (Masuda 1993).

H. Natural Surveillance

Householders may trim bushes at the front of their homes, and banks may light the interior of their premises at night in attempts to capitalize on the "natural" surveillance provided by people going about their everyday business. Enhancing natural surveillance is a prime objective of improved street lighting (Tien et al. 1979; Ramsay 1991*a*), of defensible space (Mayhew 1979; Coleman 1985), and of "neighborhood watch" (Rosenbaum 1988; Bennett 1990). Despite their popularity, none of these strategies has been shown to be highly effective, and, even if some successes can be documented, the capacity of "natural surveillance" to prevent crime may have been overestimated.

Improved lighting of the exterior of stores was one component of a successful program that significantly reduced burglary on a commercial strip in Portland, Oregon (Griswold 1984).

Enhancement of lighting in four badly lit and crime-prone streets in London has been followed by substantial reductions in crime and fear

of crime as well as increases in night-time pedestrian traffic (Painter, forthcoming).

Components of successful robbery prevention in convenience stores in Florida included an unobstructed view of the store's interior from outside and location of stores near evening commercial activity (Hunter and Jeffery 1992).

An "apartment watch" program combined with target hardening achieved an 82 percent reduction in reported burglaries in four apartment blocks in Ottawa (Meredith and Paquette 1992).

I. Target Removal

Some of the best examples of target removal come from attempts to deal with attacks on public telephones in Great Britain. Because the kiosk itself (especially the glass) is much more frequently vandalized than the phone, kiosks in high-risk locations have been replaced by booths. In addition, the smaller, highly vulnerable, glass panes in earlier kiosks have been replaced by larger panes in more recent designs (Markus 1984). A third example is provided by the introduction of the Phonecard, which by dispensing with the need for public telephones to store large sums of cash has removed an important target for theft.

Moore (1987) has described a highly successful theft prevention program that consisted of persuading inpatients to surrender their valuables for safekeeping or not to bring them to the hospital.

Pease (1991) has shown that a package of measures to prevent repeat victimization of houses on a public housing estate in Britain, including the removal of gas and electric coin meters that were frequent targets for theft, reduced burglaries on the estate from 526 in the year before intervention to 132 three years later.

A variety of cash reduction measures, including the use of safes with time locks, progressively introduced during the 1980s in betting shops in Victoria, Australia, substantially reduced robberies. Not only did the number of robberies decline but the average take of "successful" robberies was also substantially reduced. A cost-effectiveness calculation suggested that the measures had more than paid for themselves (Clarke and McGrath 1990).

Chaiken, Lawless, and Stevenson (1974) showed that bus robberies in New York City dropped from a high of sixty-seven per month just prior to the introduction of exact fares to seven or less per month afterward. A similar result was also reported for eighteen other cities

that introduced exact fare systems for buses (Stanford Research Institute 1970).

J. Identifying Property

The most developed programs of identifying property relate to vehicles. Registration of motor vehicles was required in some U.S. states from almost the beginning of the century, and, subsequently, all vehicles sold in the United States were required to carry a unique Vehicle Identification Number. More sophisticated forms of property marking recently deployed against auto theft include PINs for car radios and LOJACK, the trade name for a system involving the concealment of a small transmitter in the body of an automobile that facilitates its recovery if stolen (Clarke and Harris 1992*a*).

Illinois was one of the last U.S. states to require vehicle registration (in 1934), whereupon vehicle thefts declined from 28,000 in the previous year to about 13,000 (Hall 1952).

Laycock (1991) provides evidence that property marking undertaken in three small communities in Wales, combined with extensive media publicity, nearly halved the number of reported residential burglaries, with 66 in the second year of the program compared with 128 in the year before. Theft rates of automobiles fitted with security-coded radios that cannot be operated without a PIN have recently declined in Australia (National Roads and Motorists' Association NRMA Insurance Ltd. 1990).

K. Removing Inducements

The mere presence of a weapon, such as a gun, has been found to induce aggressive responses in some people (Berkowitz and LePage 1967). This evidence suggests that, apart from reducing availability of weapons, gun controls would also remove inducements to violence. Further examples of reducing inducements can be found in Wise's (1982) suggestions for a "gentle deterrent" to vandalism, such as painting murals on the bare surfaces of walls that almost invite graffiti, and the so-called bum-proof benches at bus stops designed to prevent their use as beds by alcoholics and itinerants (Davis 1990). Arguments in favor of the rapid repair of vandalism, on the grounds that damaged objects invite further attack, also fall under the heading of removing inducements. Support for these arguments is provided by Zimbardo's (1973) finding that a car left parked in poor condition in an inner-city area rapidly attracted further depredation and Samdahl and Christian-

sen's (1985) finding that picnic tables that had been scratched and carved were much more likely to be damaged further than tables not so marked. The arguments have been taken a step further by Wilson and Kelling (1982) who have maintained that the failure to deal promptly with minor signs of decay in a community, such as panhandling or soliciting by prostitutes, can result in a quickly deteriorating situation as hardened offenders move into the area to exploit the breakdown in control.

Sloan-Howitt and Kelling (1990) have documented the remarkable success achieved within five years by the New York Transit Authority in ridding its 6,245 subway cars of graffiti, an important component of which was a policy of immediate cleansing. This removed the gratification for offenders of seeing their work on public display.

Substantially reduced rates of crime and vandalism on the metropolitan railways of Victoria, Australia, were reported by Carr and Spring (1993) following the introduction of Travel Safe, a program consisting of enhanced security and rapid repair of vandalism and graffiti. For example, the weekly number of train windows needing repair progressively declined from 372 in October 1990 to 105 in May 1991.

L. Rule Setting

All organizations find it necessary to have rules about conduct in their fields of jurisdiction. For example, most businesses regulate employees' telephone use, and all retail establishments require employees to follow strict cash-handling and stock control procedures. Organizations such as hospitals, schools, parks, transportation systems, and hotels and restaurants must, in addition, regulate the conduct of the clienteles they serve. Any ambiguity in these regulations will be exploited where it is to the advantage of the individual. (Most attempts to avoid income tax relate to those sections of the Internal Revenue Service tax return that are more difficult to investigate [Klepper and Nagin 1987].) One important strand of situational prevention, therefore, is the introduction of new rules or procedures (and the improvement of those in place) that are intended to remove any ambiguity concerning the acceptability of conduct. The existence of these rules means that offenders must be prepared to incur higher costs in terms of fear or conscience. Such rules would not normally require the backing of the law, but some highly specific laws and local ordinances have the character of situational measures.

In an attempt to produce consensual crowd management at the Aus-

tralian Motorcycle Grand Prix, riders who were permitted to operate campsites for their fellow motorcyclists were encouraged to develop rules and procedures for use of the facilities. This was one element of a package of situational measures that produced a largely trouble-free event in 1991 at Philip Island, Victoria. In previous years, the event had been marred by brawls between the police and the crowds (Veno and Veno 1993).

Introduction of a widely supported local ordinance banning the consumption of alcohol in the streets and open public spaces of central Coventry was followed by reductions of about 40 percent in the number of complaints of insulting behavior and of more than 50 percent in the numbers of people who regarded public drinking as a common problem in Coventry (Ramsay 1991*b*).

M. "Everything Works"?

Cataloging "successes" as in this section, even while acknowledging that many are vulnerable to methodological criticism, may seem to suggest that in situational prevention "everything works," when this is patently not the case. First, impressive as are the reductions in crime (which were sometimes 50 percent or more), it is also clear that, at best, situational measures have ameliorated rather than cured the problem. Second, in some cases discussed in Section IV, the main result of situational measures was to displace not cure the problem. Third, there are numerous instances where situational measures have proved vulnerable to implementation difficulties. In some cases (examples are provided in Section V and in the essay by Laycock and Tilley [in this volume]) these difficulties have involved genuine dilemmas of ethics or costs. In others, the failures have been the result of technical or administrative ineptitude, as when anticlimb paint to deter school break-ins was too thinly applied (Hope and Murphy 1983) or when a scheme to defeat vandalism by repairing particular windows with toughened glass when they were broken proved too complicated for school maintenance staff to administer (Gladstone 1980).

Finally, there are other cases where situational measures have been implemented as planned, but which have not proved successful. This has been for a variety of reasons, as follows.

First, some measures have been too easily defeated by offenders, as in the case of the early steering locks in the United States and Britain which proved highly vulnerable to slide hammers (Clarke and Harris

1992*a*) and that of the "smart" credit cards in France which could be disabled by stamping on the chip (Levi 1992).

Second, too much vigilance has sometimes been assumed on the part of guards or ordinary citizens. Security guards rarely monitor CCTV systems as closely as expected by the designers; people pay far less attention to what is going on outside their homes than is sometimes assumed by neighborhood watch schemes and defensible space designs (Mayhew 1979); and people rarely respond to car alarms, so that the main result of their widespread deployment has been to reduce further the quality of life in cities (Clarke and Harris 1992*a*).

Third, measures have occasionally provoked offenders to unacceptable escalation as in the case of the bulletproofing of token booths on the New York subway which resulted in some attacks on booths with gasoline-fueled fires (Dwyer 1991).

Fourth, some measures have facilitated rather than frustrated crime: Ekblom (1991) cites the example of pickpockets on the London Underground who stationed themselves near signs warning of theft to see which pockets were checked by passengers on reading the signs; and one result of introducing traffic barriers in Vancouver to frustrate cruising johns was that the prostitutes could sit on the bollards and proposition clients who had been forced to slow down (Lowman 1992).

Fifth, in other cases, measures have been defeated by the carelessness or idleness of potential victims. Residents routinely frustrate entry control systems on apartment buildings by propping open the doors to save themselves from answering the door or from taking a key when leaving on a short errand. The preventive value of the early security-coded radios in the United States was reduced because car owners did not enter their private codes, thus allowing the radios to revert to a standard code known to thieves (Braga and Clarke, forthcoming).

Sixth, another reason for the initial lack of success of these radios was that some thieves did not know which radios were security-coded and continued to steal some they could not use. This flaw was remedied by the introduction of a continuously blinking light on the radio indicating that it was security-coded (Braga and Clarke, forthcoming).

Seventh, some inappropriate measures have been introduced because no proper analysis of the problem was undertaken. For example, Harris and Clarke (1991) have argued that the parts-marking provisions of the federal Motor Vehicle Theft Law Enforcement Act of 1984 are doomed to failure because parts marking was restricted only to "high

risk" automobiles; leaving aside the resultant scope for displacement, most of the defined "high risk" models are taken not for chopping, but for joyriding, which will not be deterred by parts marking. Measures to disrupt prostitution in Vancouver resulted largely in displacement of the problem because, unlike the case in a London suburb where similar measures had been successful, the prostitutes involved were highly dependent on their earnings to support drug habits or for other reasons (Lowman 1992).

Eighth, other measures have proved unsuitable because insufficient thought has been given to users' needs. For example, one security innovation left senior citizens "trapped inside a fortress of heavy doors and electronic card-key devices which they found difficult to understand and to operate, while neighbours were no longer able to keep a friendly eye on them" (Sampson et al. 1988, p. 484).

As these examples make clear, situational measures do not always work in the intended ways. It is also the case that measures that work in one setting may not do so in another, and not just for the reason (as in the prostitution example above) that different groups of offenders are involved. Thus helmet-wearing laws that serendipitously but dramatically reduced motorcycle theft in Germany (see the next section) appear to have had much less effect in the United States. This is because in the United States the laws were not universally applied as in Germany but were introduced in a piecemeal and inconsistent fashion. Thus laws were introduced in some states, but not in neighboring ones; when in force, laws sometimes applied only to certain sections of the population, for example, those aged below twenty-one, and not to others; and in some states helmet laws were successively introduced, repealed, and introduced again. The result has been confusion and lack of public support. This has meant that the laws have been widely disregarded, which has destroyed their serendipitous value in reducing motorcycle theft.

None of the failures of situational prevention seriously call into question the basic validity of the concept, but they do suggest that matters are more complex than those implementing situational measures frequently appreciate. Much detailed thought has to be put into the design and implementation of measures. They have to be carefully tailored to the settings in which they are applied, with due regard to the motives and methods of the offenders involved, to the costs and acceptability of the solutions proposed, and to the possible unwanted consequences in terms of displacement or escalation. In many cases, those seeking

to prevent crime will be competing with inventive and determined offenders trying to defeat the measures introduced. The task is not an impossible one, however, as shown by the many examples of success.

The challenge for research is to help practitioners avoid the pitfalls by providing a sounder base of knowledge on which to act. All we know at present is that some measures work well in certain conditions. What we need to know is which measures work best, in which combination (Tilley 1993*b*), deployed against what kinds of crime and under what conditions (Poyner 1993). This will require a greatly increased investment in evaluative research, making use of more complex methodology than has often been employed in the past.

IV. Displacement and Diffusion

Under the dispositional assumptions of traditional criminological theory, situational variables merely determine the time and place of offending and manipulating situations would simply cause offenders to shift their attention to some other target, time, or place; change their tactics; or even switch to some other categories of crime (Reppetto 1976). This seeming inevitability of displacement has constituted the principal theoretical criticism of situational prevention, but this has changed with the theoretical developments described in Section I. Under the rational choice assumptions that now guide thinking about situational prevention, displacement is no longer seen as inevitable but as contingent on the offender's judgments about the ease, risks, and attractiveness of alternative crimes. If these alternatives are not viable, it is entirely conceivable that the offender may settle for smaller criminal rewards or for a lower rate of crime. Very few offenders are so driven by need or desire that they have to maintain a certain level of offending whatever the cost. For many, the elimination of easy opportunities for crime may actually encourage them to explore noncriminal alternatives. However, since crime is regarded under the rational choice perspective as the product of purposive and sometimes inventive minds, displacement to apparently quite different categories of offense would not be unexpected, so long as these new crimes served the same purposes as the ones that were thwarted.

In this section, evidence about displacement is reviewed, and the implications of the rational choice perspective are more fully explored. The section closes with a discussion of the more recently recognized phenomenon of "diffusion of benefits." Like displacement, this is a consequence of situational measures, but, since it refers to unexpected

reductions in crimes that have not been targeted by the measures, one that has positive implications.

A. Displacement of Crime

Numerous examples of displacement have been reported, particularly in the earlier literature (Gabor 1990). Street crimes increased in surrounding districts following a successful crackdown on these crimes in one New York City precinct (Press 1971). The reduction in robberies following the introduction of exact fare systems on New York City buses was accompanied by an increase in robbery in the subway (Chaiken, Lawless, and Stevenson 1974). In Columbus, Ohio, a police helicopter patrol (Lateef 1974) and, in Newark, New Jersey, a street-lighting program (Tyrpak 1975) appeared to shift crime to precincts not covered by the new measures. The reduced risk of auto theft for new vehicles fitted with steering column locks in Britain was found to be at the expense of an increased risk for older vehicles without the locks (Mayhew et al. 1976). Gabor (1981) found in Ottawa that a property-marking program may have displaced burglaries from the homes of participants to those of nonparticipants. Finally, Allatt (1984) found that the decrease in burglary on a British public housing estate which had undergone a program of "target hardening" was accompanied by an increase in property crimes in adjacent areas.

Apart from these and other instances in which displacement was found, it has often been claimed that researchers have failed to detect displacement that has in fact occurred. This would be especially likely where the displacement involved kinds of crimes other than the ones targeted. Thus the success achieved in reducing aircraft hijackings in the 1970s by baggage screening might possibly have led to an undetected increase in other terrorist activity, such as car bombings, assassinations, and hostage taking. The methodological difficulty encountered in detecting such displacement was explained in an earlier *Crime and Justice* discussion, as follows: "Some displaced crime will probably fall outside the areas and types of crime being studied or be so dispersed as to be masked by background variation. . . . The wider the scope of the study in terms of types of crimes and places, the thinner the patina of displaced crime could be spread across them; thus disappearing into the realm of measurement error" (Barr and Pease 1990, p. 293).

The uncritical acceptance of displacement, however, may also mean that increases in crime, which might have occurred anyway, have sometimes been wrongly attributed to displacement. It has recently

been shown that the appearance of a new form of slug soon after the ticket machines of the London Underground had been modified to prevent the use of an earlier, more primitive slug was unlikely to have been due to displacement even though this was believed to be the case by the officials dealing with the problem. Clarke, Cody, and Natarajan (1994) demonstrated that the distribution of the new slugs across the stations of the London Underground was quite different from that of the earlier slugs. This suggested that different groups of offenders were responsible for the new slugs, and it was concluded that, even if action had not been taken against the original slugs, the new ones might still have appeared.

Moreover, in none of the instances of observed displacement did it seem to be 100 percent (Gabor 1990), and, with the development of rational choice analyses, evidence has begun to accumulate of the successful application of situational measures with apparently few displacement costs. Much of this evidence relates to the examples of success cited in Section III. For instance, Knutsson and Kuhlhorn (1981) could find no evidence of an increase in a range of "conceivable" alternative crimes following the implementation of new identification procedures in Sweden that greatly reduced check frauds. Following the rearrangement of market stalls and improved lighting which reduced thefts at covered markets in Birmingham, Poyner and Webb (1987) found no evidence of displacement of thefts to other nearby markets. There was little evidence that reductions in obscene phone calling in areas of New Jersey where Caller-ID became available had led to an increase of such calls in other areas (Clarke 1990). Finally, Clarke, Field, and McGrath (1991) showed that reductions in bank robbery in Australia brought about by target hardening were not followed by increases in armed robberies of alternative targets such as credit unions, betting shops, convenience stores, and service stations.

In other cases cited in Section III, the nature of the targeted offenses would have meant there was little point in looking for displacement. For example, it is unlikely that those deterred by random breath tests from drunken driving in New South Wales (Homel 1993) or those deterred by speed cameras from speeding in Victoria (Bourne and Cooke 1993) would have displaced these behaviors to some other time or place. People do not usually set out to commit these offenses but will do so when circumstances dictate (Homel 1993). One important circumstance is the perceived chance of arrest and, were the ubiquitous

speed cameras or random-breath-testing patrols to be withdrawn, people would no doubt once again revert to their old ways.

Once the contingent nature of displacement is recognized, it is not usually difficult to explain why it may not have occurred in particular instances. For example, Clarke, Field, and McGrath (1991) explained the lack of displacement of armed robberies from target-hardened banks to convenience stores and petrol (gas) service stations on the grounds that these latter targets would not generally yield such large "takes" and might be beneath the dignity of the more organized criminal gangs. In explaining the apparent lack of displacement of obscene phone calls to other areas following the introduction of Caller-ID in parts of New Jersey, Clarke (1990) points out that by no means all obscene phone callers are persistent random dialers hoping to hit on a susceptible woman. Many appear to victimize only particular women of their acquaintance, and it is unlikely that, with the introduction of Caller-ID in their local telephone areas, these individuals would have begun to call to more distant parts of New Jersey where they would be less likely to know anyone. (Incidentally, as drinking and masturbation seem to be quite frequent accompaniments of obscene phone calling, it is unlikely that calling from a public phone box to avoid the risk of identification would often be a practicable alternative to calling from one's home or place of business.) Finally, in explaining why prostitutes did not simply move to other locations following successful action to close down the red-light district in Finsbury Park, Matthews (1990) cites the low commitment to prostitution of many of the women involved, who appeared to be drawn to Finsbury Park by its reputation as a good place to solicit. Interviews with some of these women revealed that many saw prostitution as a relatively easy way to make a living but on occasion were employed in other unskilled occupations, working also as waitresses or shop assistants. As the environment in Finsbury Park became less hospitable, "It would seem that over a period of about one year, most of the girls gave up prostitution or moved back home or elsewhere. For many, their normal period of involvement in prostitution may have been three or four years and, therefore, the effect of intensive policing was to shorten that period for a year or two in most cases" (Matthews 1990, p. 186).

Deeper understanding of the motives and modus operandi of target groups of offenders, as obtained in Matthews's study, provides a way of dealing with the limitations of the statistical search for displacement

discussed by Barr and Pease (1990). It may not always be possible to interview offenders, but in some cases insights into motivation and methods can be provided by closer analysis of patterns of offending. For example, Clarke and Harris (1992*b*) have shown important differences among automobiles in their risks for different forms of theft, which reflect the motives of offenders. Thus new cars most at risk of "stripping" in the United States during the mid-1980s were predominantly European models with good audio equipment, those most at risk of "joyriding" were American-made "muscle" cars, and those at risk of theft for resale were mostly higher-priced luxury automobiles. These "choice structuring properties" (Cornish and Clarke 1987) of the target vehicles are not difficult to understand in terms of the motives of offenders and would also help to direct the search for displacement if the security were improved for any subset of vehicles. Thus if some "muscle" cars were made more difficult to take for joyriding, it would make sense to confine the search for displacement only to others of the same group.

Similar logic was followed by Mayhew, Clarke, and Elliott (1989) in their study of displacement following the reduction of motorcycle thefts in West Germany between 1980 and 1986, brought about by the progressive enforcement of helmet legislation. During this period motorcycle thefts declined by more than 100,000 because the helmet requirement substantially increased the risks of opportunistic thefts for those offenders who were unable at the same time to steal a helmet. Mayhew and her colleagues reasoned that since many opportunistic thefts would have been for purposes of joyriding or temporary use (e.g., to get home late at night), the most likely result of the reduced opportunities for stealing motorcycles would be an increase in thefts of cars and bicycles. In fact, as shown by the data in table 2, there was little evidence of displacement to either category of target. Car thefts did rise over the same period, but only by a few thousand, while bicycle thefts declined (after an initial rise) to below their previous level. Bicycles may not usually provide a realistic or attractive alternative means of transportation, whereas cars may not provide the same joyriding thrills, may require more knowledge to operate, and may be more difficult to steal.

A final illustration of the value of considering "choice structuring properties" is provided by the British gas suicide story (Clarke and Mayhew 1988). The elimination of gas suicides in Britain in the 1960s

TABLE 2

Thefts of Motorcycles, Cars, and Bicycles: Federal Republic of Germany, 1980–86

Year	Motorcycles	Cars	Bicycles
1980	153,153	64,131	358,865
1981	143,317	71,916	410,223
1982	134,735	78,543	453,850
1983	118,550	82,211	415,398
1984	90,008	72,170	376,946
1985	73,442	69,659	337,337
1986	54,208	70,245	301,890

SOURCE.—Mayhew, Clarke, and Elliott (1989).

and 1970s resulting from the introduction of natural gas (which contains no toxins) was not followed by substantial displacement to other forms of suicide. Consequently, the overall suicide rate for the country declined by about 40 percent. The lack of displacement was explained by Clarke and Mayhew in terms of the particular advantages of the use of domestic gas as a form of suicide. It was readily available in every home, it was simple to use, and it was highly lethal. It was also painless, left no marks or blood, and required little courage. No other alternative possessed all these advantages and would therefore have provided an acceptable alternative for many people. Sleeping pills and other poisons were much less lethal. Easy access to car exhaust gas was limited to a minority of the population. Guns and knives were perceived as violent and painful. In Britain, hanging is perceived to be a death reserved for traitors and murderers. Finally, most older people would find it difficult to clamber up steep cliffs or down to railway tracks.

It has been argued that it will only be a matter of time before the suicidally inclined displace to other methods and the suicide rate reverts to its former level; indeed, this is now true of the suicide rate for males. However, there is substantial evidence that for many people the urge to kill themselves is in response to situational stress (such as a bereavement) and may dissipate as depression is alleviated. It is therefore unlikely that the suicide rate has been substantially swelled in recent years by individuals who were prevented from killing themselves in earlier years by the detoxification of gas but who eventually

succeeded with another method. The gradual increase in the male suicide rate may reflect, not delayed displacement, but underlying motivational trends. Even if there has been an increase in the use of other methods (including exhaust gases [Clarke and Lester 1987]), any such changes might have occurred anyway, even if poisoning by domestic gas were still available as a method.

This argument about the consequences of gas detoxification should not be taken to mean that longer-term adaptations do not occur in response to situational measures. Indeed, it is highly probable that such changes do occur. For example, it has frequently been claimed that car thieves have found ways around steering column locks, and, indeed, there is some support for this in that rates of car theft in some countries where these locks have been fitted for many years have not declined as much as expected (Clarke and Harris 1992*a*). However, it is too easy to assume that offenders will eventually always find a way around preventive measures. Even in the case of steering column locks, the experience has not always been consistent with this view. It has recently been shown that the reduction in the rate of car thefts in West Germany, brought about by the introduction of steering column locks in the 1960s, has persisted to this day (Webb and Laycock 1992; Webb 1994). In contrast, the rate of theft from vehicles (which steering column locks have little impact on) has increased steadily over the past thirty years (see fig. 2).

Clarke and Harris (1992*a*) have argued that a possible reason for the enduring success of the steering locks in West Germany (apart from the possibly higher quality of the German equipment) is that, because the locks were made compulsory for all cars on the road, theft was greatly reduced almost overnight. The long-term consequence of this was that neophytes were frequently deprived of the example and tutelage of experienced car thieves, which may have helped to destroy local "car theft cultures" in many parts of the country. This affords a marked contrast with the situation in most other countries (including the United States and Britain) where the locks were introduced only for new cars at manufacture. Consequently, car thieves could continue to operate so long as they concentrated their efforts on older vehicles. They could also gradually learn ways of defeating the steering locks and could continue to pass on the tricks of the trade to novices. This may help to explain why rates of car theft in these other countries, even though most cars on the road now have the steering column locks, are higher than in Germany (Webb 1994).

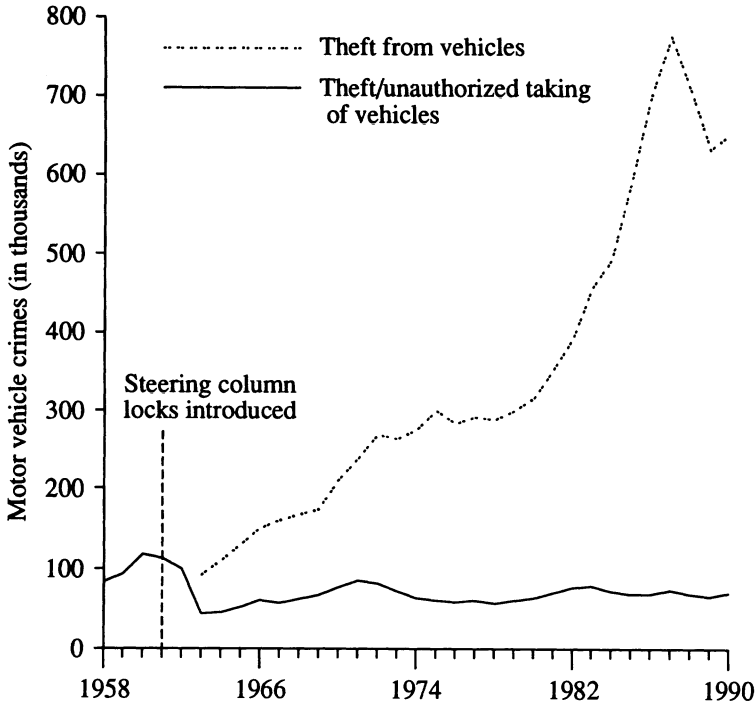

FIG. 2.—Motor vehicle crime in the Federal Republic of Germany, 1958–90. Source: Webb and Laycock (1992).

B. Diffusion of Benefits

It has been argued above that the rational choice perspective results in a more complex, and ultimately more optimistic, view of displacement. While the possibility of displacement should never be overlooked, and while it may often be difficult to detect, it is clearly not as inevitable as some of the earlier dispositional thinking assumed. Moreover, as argued by Barr and Pease (1990), even when it occurs it may sometimes be "benign" as in the case, for example, of preventive measures which bring relief to repeatedly victimized groups although at the cost of an increased risk for others. This observation provided the rationale for decisions about the allocation of scarce preventive resources in an experiment to reduce burglary in Kirkholt, a public housing estate in the North of England (Pease 1991). Target-hardening priority was given to houses that had recently experienced a burglary, with the result that, despite their higher risks, very few of these houses experienced a further burglary in the follow-up period. Pease also

noted that these preventive benefits permeated, through what he called a process of "drip-feed," to other households that were not target-hardened so that the burglary rate for the whole of the Kirkholt estate declined dramatically.

The "drip-feed" effect is, of course, the reverse of displacement in that preventive action led, not to an increase, but to a reduction in crimes not directly addressed by the measures. As Clarke and Weisburd (1994) observe, similar effects have been noted under a variety of other names. For example, Miethe (1991) has referred to the "free rider" effect when residents benefit from the crime prevention measures taken by their neighbors, and Sherman (1990) to the "bonus" effect sometimes observed in police crackdowns when there is a carryover of the preventive effect beyond the period when the crackdown is in force. Scherdin (1986) used the term "halo effect" when reporting that book detection systems prevented thefts, not just of the electronically protected materials, but also of other materials as well. In some cases, the phenomenon has been reported without giving it a name. Poyner and Webb (1987) found that measures to reduce thefts from shopping bags in particular city center markets seemed also to reduce thefts in other markets as well. In his evaluation of a CCTV system installed to reduce auto theft at a university, Poyner (1991) found an equal reduction of crime in the parking lot not covered by cameras as in the ones covered. In his study of CCTV on buses, he found that damage and other misbehavior was reduced, not only on five buses fitted with cameras, but throughout the whole fleet of eighty buses (Poyner 1988).

Despite the variety of terminology, in all these cases the same phenomenon has been observed. That is to say, reductions in crime have occurred which are difficult to attribute to the direct action of situational measures. Clarke and Weisburd (1994) have argued that the generality of the phenomenon demands a standard term and have proposed "diffusion of benefits," since the geographical and temporal connotations of this term parallel those of "displacement of crime." They have defined diffusion as "the spread of the beneficial influence of an intervention beyond the places which are directly targeted, the individuals who are the subject of control, the crimes which are the focus of intervention or the time periods in which an intervention is brought" (Clarke and Weisburd 1994, p. 169). They have also distinguished between two forms of diffusion which they call "deterrence" and "dis-

couragement." Deterrence was invoked by Scherdin (1986), for example, in explaining why the book detection system she studied also prevented thefts of items that were not electronically tagged and by Poyner in identifying reasons for the general decline in damage to the fleet when only some of the buses were fitted with CCTV cameras: "The children have learned . . . that the cameras will enable misbehaving individuals to be picked out and that action will be taken. . . . They appear to believe that most buses have cameras, or at least they are uncertain about which buses have cameras" (Poyner 1988, p. 50).

Deterrence affects offenders' assessments of risk, but, for diffusion by "discouragement," the key is not the judgment of risk, but the assessment of effort and reward. For example, one component of the successful action against burglary in Kirkholt was the removal of prepayment meters from many houses on the estate (Pease 1991). This seems to have been enough to discourage potential burglars, who could no longer be sure of finding a meter containing cash without expending much additional effort. Similarly, the drop in thefts at all Birmingham city center markets following the situational measures taken at only some of them may have been due to the fact that "the general attractiveness of this area for thieves has reduced" (Poyner and Webb 1987). Ekblom (1988) accounted for the finding that antibandit screens in London post offices had produced a reduction, not just in over-the-counter robberies, but also in other robberies of staff and customers when he speculated that would-be robbers may have received "the very general message that something had been done to improve security at the sub-post offices" (Ekblom 1988, p. 39). Finally, Clarke, Field, and McGrath (1991) have suggested that an intensive target-hardening program in Australian banks brought about a general reduction in robberies of all commercial targets (including convenience stores, gas stations, and betting shops) because robbers began to believe that this form of crime was no longer worth pursuing.

It would be difficult to overestimate the importance of diffusion of benefits if the phenomenon is as common as these various examples suggest. At the very least, the possibility of diffusion will have a profound effect on the design of evaluations, particularly those that make use of "control" areas not directly targeted by situational measures and supposedly unaffected by them. Indeed, there may already have been evaluations in which the benefits of situational action have been under-

estimated because falls in crime were also observed in "control" areas, with the conclusion that both "experimental" and "control" areas had been subject to a general but extraneous decline in crime. Whatever the case previously, it is clear that in the future, evaluators must pay attention not only to the problems of displacement but also to the benefits of diffusion.

It is also clear that much more needs to be discovered about ways of enhancing diffusion. Sherman (1990) has suggested that the "free bonus" of crackdowns might be increased by randomly rotating crackdowns and back offs across times and places so as to lead offenders to overestimate the actual levels of risk in force on any particular occasion. He also advocated the deliberate use of publicity about imminent crackdowns to promote this uncertainty in offenders' minds. Clarke and Weisburd (1994) argue that these strategies might be employed to diffuse the benefits of forms of "situational deterrence" (Cusson 1993) other than crackdowns, and they also identify other possible means of enhancing diffusion, such as concentrating preventive action on highly visible or attractive targets (i.e., Pease's [1991] "drip feeding" of crime prevention), so as to lead offenders to think that preventive measures may have been more generally applied.

Since these various strategies depend on influencing judgments made by offenders, we need to learn more about ways that offenders obtain and process information about preventive initiatives and what role is played in this process by their own direct observation, their relationships with other offenders, and information obtained through the media. Such research would be of broader theoretical interest if it were to be conducted within a larger program of work on how offenders perceive and evaluate the opportunity structure for crime and how they learn about ways in which this expands as well as contracts. Whatever the practical payoff from such studies and other related research on offender perceptions, it is likely that, in the 1990s, diffusion of benefits may come to supplant displacement as the principal focus for theoretical debate about the value of situational measures.

V. Ideology and Implementation

Given the numerous policy advantages of situational prevention, it might have been expected to achieve broad acceptance across the political spectrum. The Left might have welcomed its focus on local problems and local decision making; liberals might have been attracted to its essentially nonpunitive philosophy; and conservatives might have

been attuned to its message concerning the need for agencies and communities to take the initiative in dealing with their crime problems. Instead, it has often been met with indifference and even hostility. The Right, and this includes most varieties of political opinion in the United States, sees it as a fundamentally irrelevant response to crime because it eschews questions of moral culpability and punishment (see Bright 1992). The Left criticizes it for neglecting issues of social justice and for being too accepting of establishment definitions of crime (Young 1988). Liberals fear that its avoidance of the "root causes" of crime, such as poverty, deprivation, and unemployment, may divert attention from these problems.

Much of this hostility has been fueled by the more unattractive, target-hardening forms of situational prevention which have been seen as harbingers of a "fortress society," in which people fearful of crime barricade themselves in their homes and places of work (Davis 1990). Increased experience with situational prevention may dispel some of these fears since many of the measures (such as steering column locks and parts marking of automobiles) are unobtrusive or may actually reduce fear of crime (e.g., through improved street lighting or defensible space architecture). Other measures which enhance security, such as bar coding of merchandise and central locking of automobiles, have considerable advantages of convenience.

It is ironic, as previously noted (Clarke 1992), that this unobtrusiveness and convenience feeds another fear: it may not be the fortress society that is imminent, but Aldous Huxley's "brave new world." The Orwellian specter of "Big Brother" state control, stimulated by the already widespread use of CCTV and intruder alarms, has been provoked by reports of rapidly approaching new forms of technology that will greatly expand the potential for totalitarian control, not of the iron fist but of the velvet glove (Marx 1986).

Little matter that these fears of the State's blanket adoption of situational controls are inconsistent with the piecemeal and analytic nature of the approach. Little matter also that such controls can often be applied only by particular private or business organizations. Rather, it seems likely that concerns about the erosion of privacy and civil liberties will increasingly be raised in opposition to situational measures. These concerns are discussed below. Also addressed separately below is another barrier to implementation: the difficulty of persuading managers not used to dealing with crime (particularly those in the private sector) to accept responsibility for preventive action. Discussion of a

wider range of implementation issues can be found in the essay by Laycock and Tilley (in this volume).

A. Privacy and Civil Liberties

Concerns about the erosion of privacy and civil liberties have already led to difficulties for a variety of preventive measures. The best-known example relates to gun controls which are vehemently opposed by a section of the U.S. population on grounds of their supposed infringements of constitutional liberties. This has made it difficult for many states to enact mild and sensible laws, endorsed by a majority of the population, prohibiting private ownership of so-called assault weapons or requiring background checks and waiting periods at the time of purchase.

A taste of things to come may also be provided by controversies surrounding the attempted implementation of two new forms of technology, Caller-ID and speed cameras (or "photo radar," which is the name in the United States). Caller-ID (which is a device that reveals the caller's telephone number to the person answering) has a number of crime prevention advantages, apart from providing a deterrent to obscene and harassing calls (Clarke 1990). It deters fraudulent sales, bomb hoaxes, and false fire alarms. It also increases the risks for burglars in using the phone to check for occupancy and for robbers in luring delivery drivers to a convenient location for attack.

Despite these crime prevention advantages and other benefits for subscribers (such as the maintenance of a record of all numbers calling while the phone is unattended), the introduction of Caller-ID has been subject to court battles in many parts of the United States. In a number of these suits, civil libertarians have successfully opposed introduction of the service on grounds that it constitutes an unwarranted intrusion on the privacy of telephone callers, particularly those with unlisted numbers (Temple and Regan 1991). In making the case against Caller-ID, recourse has been made to various arguments: it violates some state wiretap laws; it can reveal the telephone numbers of people such as secret service agents or battered women in refuges who have good reason to keep their whereabouts secret; it inhibits the use of informant hot lines and confidential telephone counseling by the Samaritans and others; and it may facilitate the development of unwelcome forms of telemarketing.

Solutions exist for most of these problems. For example, blocking mechanisms that prevent display of the number can selectively be pro-

vided. Most telephone subscribers, even those with unlisted numbers, approve of Caller-ID. Nevertheless, the legal arguments about callers' privacy have generally prevailed, and in a number of states Caller-ID has been prohibited or introduced only with a blocking option available to all subscribers. This latter effectively undermines many of the crime prevention benefits of Caller-ID. That this should be the general outcome of the court hearings is ironic because not so long ago anyone wishing to make a telephone call had to place it through an operator who might know the caller's identity as well as telephone number. This imposed a significant control on the telephone as a facilitator of crime—a control that was removed only by subsequent developments in technology.

The recent fate of attempts in New Jersey to implement a federally sponsored experiment in photo radar constitutes a similarly sorry tale. Despite its documented successes in substantially reducing speeding and fatal accidents (e.g., Bourne and Cooke 1993), the New Jersey State Assembly voted to ban the use of the device. The assemblyman who sponsored the legislation was quoted as fearing "the depersonalization of law enforcement," possible reductions in police manpower, and higher insurance rates for convicted speeders that might result from widespread use of the devices (*Law Enforcement News* 1992, p. 7; see also Marisco 1992).

This bodes ill for other possible applications of technology in preventing crime and controlling traffic infringements, such as the equipment that has reportedly been tested in Hong Kong permitting automatic monitoring of vehicle speeds through transmissions received from small radios located in individual cars (Marx 1986). Perhaps all these devices offend people's sense of fair play. People have to be given a sporting chance of getting away with crime, especially the ordinary everyday offenses that all of us might commit. Whatever the reason, these will surely not be isolated examples, and they suggest that, far from being willing dupes in accepting "the new surveillance" (Marx 1986), people will resist its deployment in the routine situations giving rise to most crime as soon as they suspect that they might fall victim to its impersonal (and one might say unbiased) law enforcement efficiency.

B. Costs and Responsibility

Most situational prevention to date has been undertaken in the public sector, and discussion of implementation difficulties has been focused

largely on ways of achieving the necessary coordination among local government agencies (e.g., Gladstone 1980; Hope 1985; Ekblom 1987). But it is now being increasingly recognized that much preventive action can be undertaken only by the private sector (e.g., by credit card companies, bus operators, offices, and shopping malls). Encouraging this action will involve government in developing a dialogue with a whole new set of actors (including the security industry, which may not welcome the potential threat to its role), most of whom can be expected to take a hard-nosed view in dealing with crime. They will see this as largely a police matter and will rarely be willing to "acknowledge that their property or operations are generating a substantial strain on police resources, accept that they have a duty, up to their level of competence, for the control of specific crimes, and take appropriate action" (Engstad and Evans 1980, p. 151). Acknowledging this responsibility would not only complicate the management task but could involve the expenditure of significant resources. Analyses of these costs are therefore likely to play an increasingly important role in crime prevention, and it is no accident that one of the first publications of the Home Office Crime Prevention Unit aimed at the private sector presents information (under the revealing title of *Making Crime Prevention Pay*) about initiatives taken by businesses that prevented crime and saved money (Burrows 1991).

The implementation issues involved are illustrated by the familiar example of shoplifting. This is promoted by many current retailing practices, such as self-service and displays that encourage impulse buying. The risks of the ensuing "shrinkage" are accepted by most stores, which rely on deterrence to contain the problem through the occasional arrest and prosecution of shoplifters. This results in significant costs being passed on to the criminal justice system, which of course are not borne by the retailers except in an indirect way through taxation.

These attitudes and practices are unlikely to change simply as a result of government exhortation (which runs the risk of being dismissed as "blaming the victim" [Karmen 1984]). Nor is it likely in the present climate that Pease's (1979) ideas about charging for police service when stores have failed to adopt preventive measures would provide a realistic alternative. However, declining retail profits, resulting from the combined pressure of a faltering economy and increased competition, might force a change. Thus it has recently been found, in the fierce competitive environment of discount electronics and appliance

sales, that sustained attention to the prevention of theft and fraud can significantly improve profit margins (Masuda 1992, 1993).

Greater attention to retail security is also likely to result from improved technology which permits instant credit checks (Levi, Bissell, and Richardson 1991) and tighter stock control (Hope 1991), but in every case of its application, the technology will need to pass cost-benefit scrutiny by the retailer. This scrutiny is unlikely to take account of the criminal justice costs of the failure to take preventive action, and there will be an increasing role here for studies by economists. Field's (1993) analysis of the costs of auto theft in the United States illustrates the potential value of such work. In a study that attempted to take account of a wide range of costs (including those falling on society as a whole, on motorists as a group, on the victims, and on the criminal justice system), he found that about two-thirds of the total cost of $35 per automobile in 1985 was accounted for by insurance premiums. Only a very small proportion of the total costs fell directly on victims as a whole (though some individual victims may suffer large losses). Under these conditions, Field argues, potential victims as a group have little incentive to invest in preventive devices. Nor is there much incentive for insurance companies to demand greater built-in security for automobiles because most of their profits come from investing premium income; the greater the risks of auto theft, the larger the premiums and the higher the profits.

Though Field dismisses arguments that motor manufacturers also have a vested interest in auto theft through the demand produced for replacement vehicles (Karmen 1981; Brill 1982), he argues that they too have little incentive to produce cars with any greater levels of security than the public demands. He doubts that even the argument that if motorists were to pay less for insurance, they would be prepared to pay more for new cars, would carry much weight in an industry that has traditionally been so concerned with reducing unit costs. He concludes that in the conditions of this "imperfect market," improved vehicle security will only come about as a result of government action to mandate vehicle security standards. Assuming that these prevented only about half of the current number of auto thefts, and assuming that the average life of a car is somewhat less than ten years, this should still produce savings in insurance costs of a couple of hundred dollars per car at current prices. Given the economies of scale resulting from the universal introduction of improved security, this amount

should be more than sufficient to pay for the improved security as well as to yield a net reduction in the overall costs of auto theft.

Even if details of Field's analysis can be criticized, it is still important for the fresh perspective it brings to bear on the discussion of auto theft prevention. There is little doubt that further economic analysis of the costs and benefits of crime prevention could be of great value in assisting government to find the best ways—whether involving tax incentives, publicity and persuasion, or mandated security standards—to promote preventive action in the private sector.

VI. Concluding Remarks

Although slow to gain acceptance in some academic and government circles, situational prevention is in a better position than when last reviewed for *Crime and Justice* in 1983. In particular, its theoretical and empirical base has been strengthened by a wide range of criminological research. Its applications have been extended to many problems of crime and disorder in a wide variety of contexts. The threat presented by displacement has receded with an improvement in theoretical understanding and an accumulation of empirical observations. Finally, it is increasingly recognized that focused preventive efforts may sometimes result in the "diffusion of benefits" in terms of reductions in crimes not directly addressed by the measures.

Particularly important has been the development of the rational choice perspective, which has provided a means of articulating the link between the two main groups of explanatory variables—offender and situational—and which has opened up a new line of policy-relevant research into offender decision making. As a consequence, situational prevention should become more attractive to criminologists, and its academic future ought to be secure, as least as judged by Colander's (1991) three requirements of publishability, teachability, and communicability. It has the potential to generate an almost unending stream of scholarly papers and dissertations. Its theoretical base is sufficiently broad and complex to satisfy the teaching requirements of a broad range of academic social scientists and to engage the interest of some, such as economists and geographers, whose assistance in developing situational prevention is needed. Its action research methodology and broad armory of practical techniques permits it to be communicated to the broad range of people—police, local officials, and managers in public and private agencies—who might have cause to investigate its relevance for their work.

The involvement of the private sector promises to extend the reach of situational prevention still further, and criminologists will have to become familiar with a host of behavioral settings—transportation systems, retail environments, factories, schools, and so forth—beyond the courts and the prisons. Indeed, it is not too fanciful to imagine that the wider application of situational prevention will result in criminologists being employed in a wide range of public and private agencies. If they are to be successful in these settings, however, they may need to adopt as mentors not sociologists and academics, but problem solvers such as traffic engineers and public health professionals.

The future reach of situational measures is also likely to be expanded by new technology (as it has been in the past decade by Caller-ID, photo radar, and many other devices) and, at least in the United States, by its growing convergence with problem-oriented policing. This convergence may help move it to a more central policy role, though implementation difficulties, ideological resistance, questions about displacement, and doubts about the capacity of situational measures to deal with violent and sexual offenses may continue to be impediments. Further loss of support from policy makers may result from its lack of dramatic effect on the totality of crime unless applied in the literally hundreds of settings that may be necessary. (It is already being criticized for not having reduced the crime rate when it has only been patchily applied in a comparatively few, and generally small-scale, projects.)

These various difficulties and criticisms will have to be addressed in the next generation of research. The need for a variety of "theoretical" studies has been identified in this essay, including research into the relationship between security and private space, into the supply of targets and crime facilitators, and into offender perceptions of the expansion of criminal opportunities resulting from social change and their contraction due to crime prevention activity. Such studies will help to clarify the theoretical limits of situational prevention, but they need to be complemented by more practically-oriented research into means of preventing particular crimes in particular contexts, into the limits of the various opportunity-reducing techniques, into ways of overcoming barriers to implementation, and into ways of containing displacement and enhancing diffusion. Research of this latter kind should help to deliver crime reductions in the increasingly efficient manner that seems likely to be demanded by the policy makers of the future.

Its future policy role may also depend, however, on the results of new work on developmental and community crime prevention directed at modifying criminal propensities. These approaches have broader appeal because they represent a more optimistic view of human nature. They do not manipulate or cajole people into compliance, but teach them to value conformity for its own sake. They lack only the evidence that they can be widely and effectively deployed, but were this forthcoming, they might readily eclipse situational prevention. However, if this evidence were not to emerge in any compelling form, it is unlikely that these approaches would fade from the policy scene since the desire to improve people is so strong and enduring. Rather, it may be that situational controls, sensitively and carefully designed to contain the unwanted results of the expanding personal freedom that accompanies technological and economic progress, will increasingly become accepted as an indispensable part of policy.

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